# Bhubaneswari Opencast Mine

## 🏭 Introduction

The **Bhubaneswari Opencast Mine** is a major coal mining project operated by Mahanadi Coalfields Limited, a subsidiary of Coal India Limited. It is located in the Talcher coalfield and is one of the important high-capacity opencast mines in India.

The mine primarily produces **non-coking thermal coal**, which is widely used in power generation. With a production capacity of around **20 million tonnes per annum (MTPA)**, Bhubaneswari plays a crucial role in supplying coal to nearby thermal power plants. It is a highly mechanized mine using modern equipment such as shovels, dumpers, and drills, making it significant for large-scale surface mining studies.

## 🌍 Geographical Overview – Bhubaneswari Opencast Mine

The Bhubaneswari mine is located in the **Angul district** of Odisha, within the **Talcher coalfield**, one of the richest coal-bearing regions in India.

### 📍 Location & Setting

- Situated in the **Talcher region** of Angul district
- Part of the **Mahanadi basin coalfields**
- Well connected by rail and road networks for coal transportation

### 🌅 Physical & Geological Features

- Lies in a **plateau region with gently undulating terrain**
- Contains **thick and extensive coal seams**, ideal for opencast mining
- Coal type: **non-coking thermal coal**

### 🌊 Nearby Natural Features

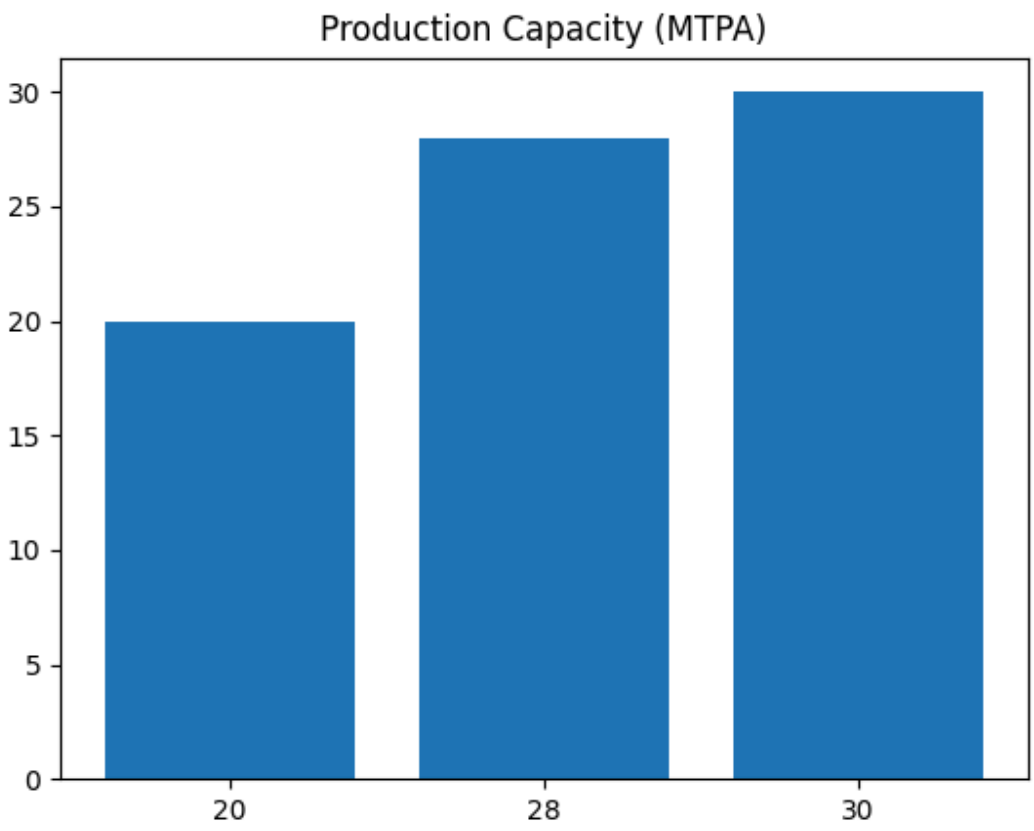
- Located near the Brahmani River basin
- Close to major thermal power plants in the Talcher region

### 🌦️ Climate

- Tropical climate
- Hot summers (up to ~45°C)

- Moderate to heavy monsoon rainfall
- Mild winters

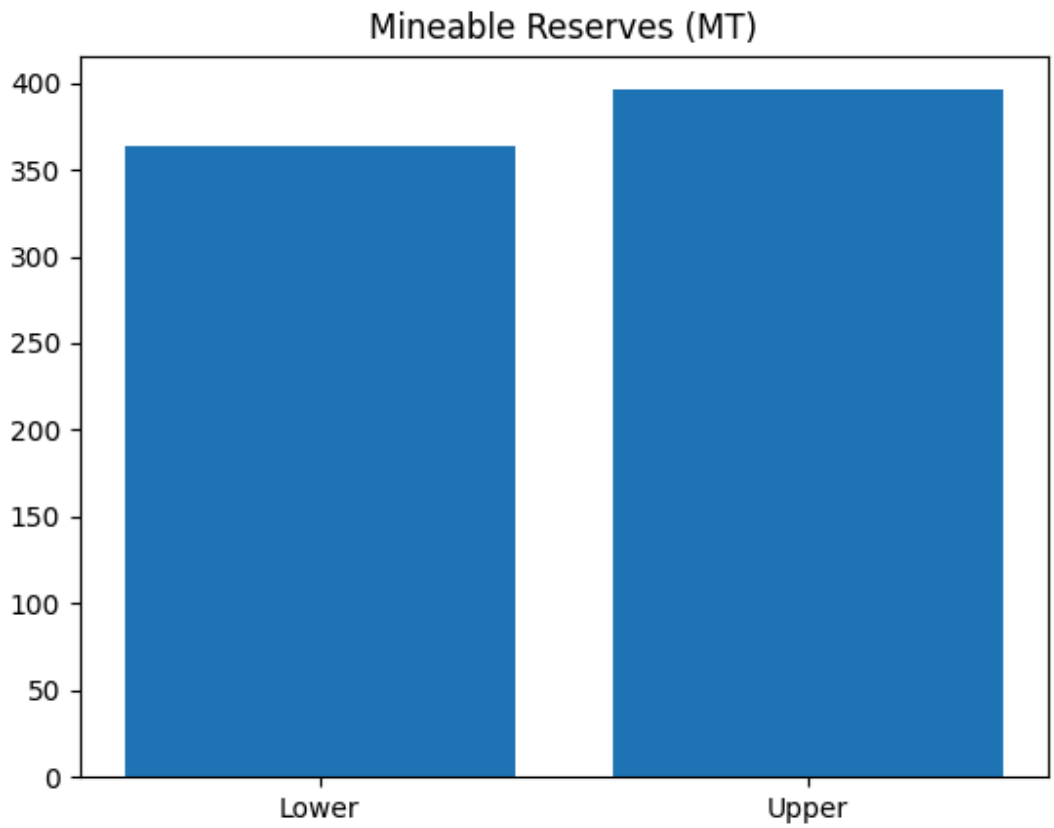
## 📊 1. Production Capacity (MTPA)




This chart shows production expansion from **20 → 28 → 30 MTPA**.
(environmentclearance.nic.in)
The mine has undergone capacity upgrades to meet rising coal demand.
**Significance:** Important for understanding mine expansion strategy and national energy contribution.

---

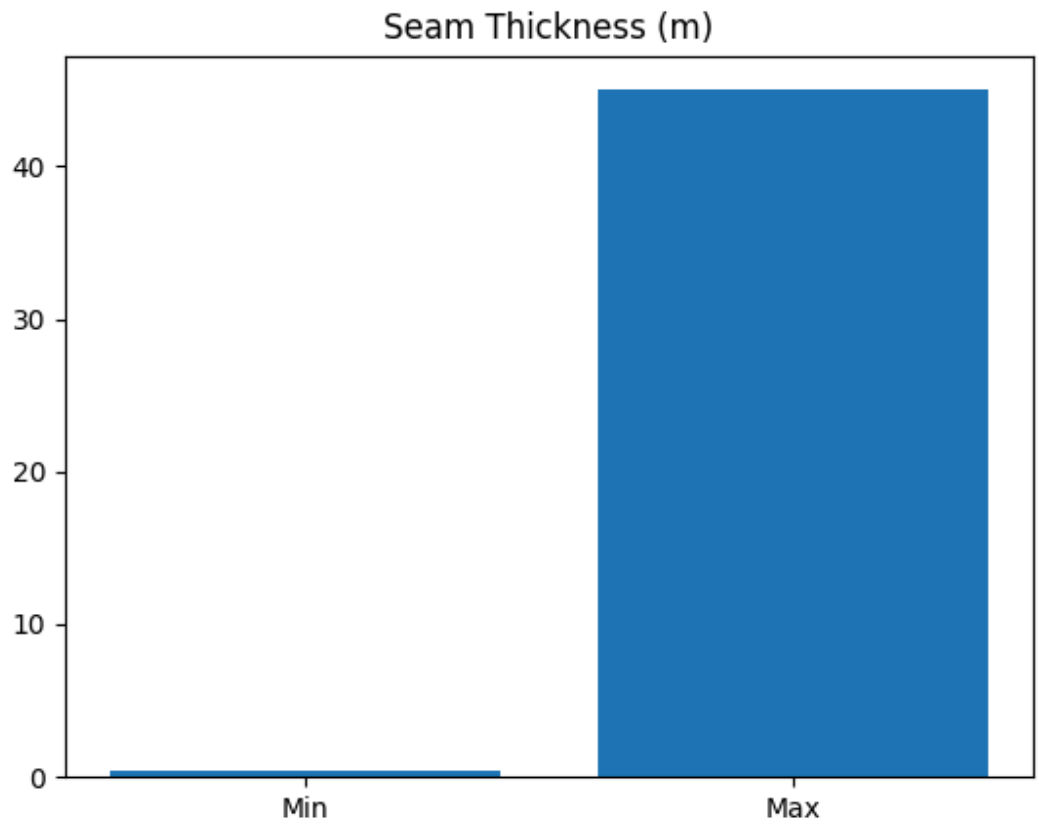
## 📊 2. Coal Reserves (Million Tonnes)




Represents **~364–396 MT mineable reserves** available in the mine. (Global Energy Monitor)
This indicates long-term sustainability of mining operations.
**Significance:** Helps estimate mine life and long-term planning.

---

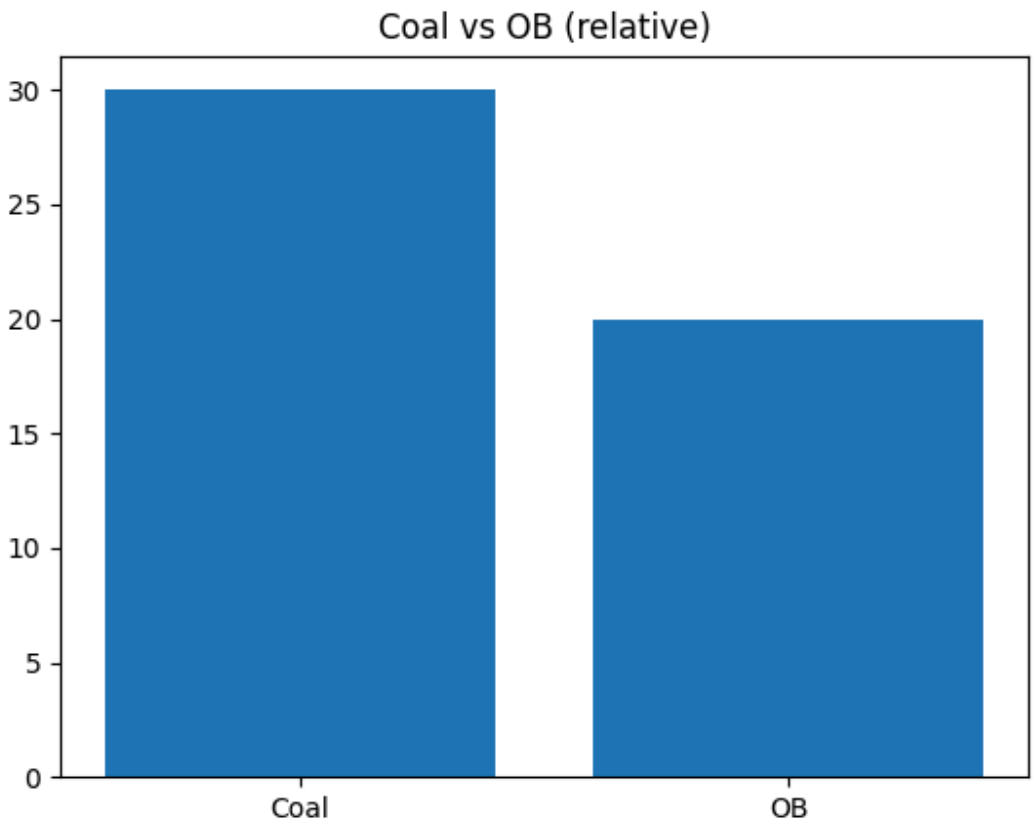
## 📊 3. Seam Thickness Distribution




Represents seam thickness ranging from **~0.42 m to ~45 m**. (environmentclearance.nic.in)
Thick seams are highly favorable for opencast mining.
**Significance:** Helps in equipment selection and mining method design.

---

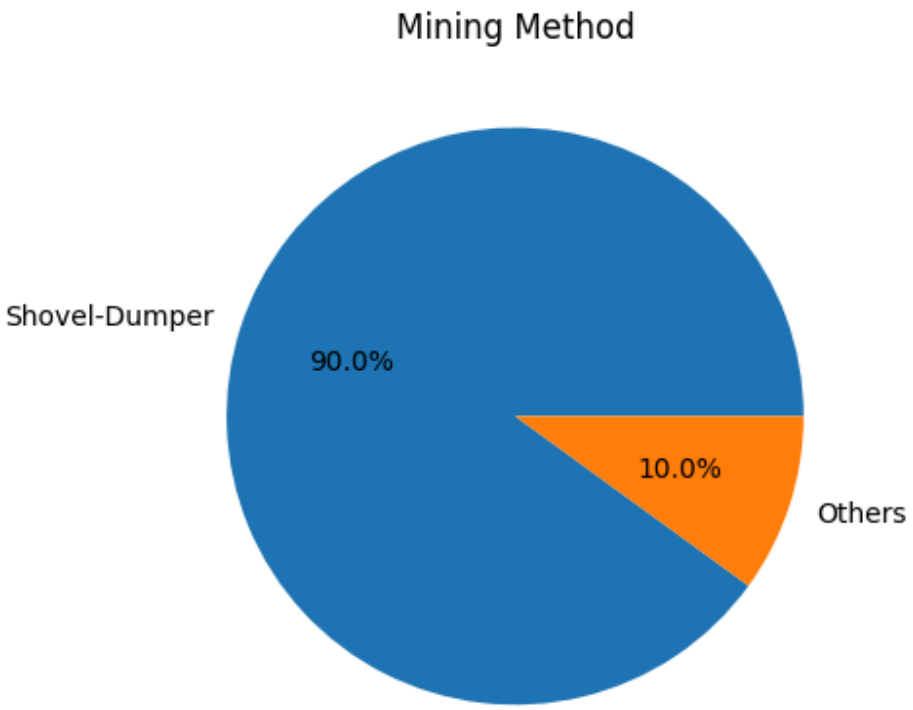
## 📊 4. Overburden vs Coal




Compares coal output with overburden removal.
Even with low SR, OB handling remains significant.
**Significance:** Important for dump planning and haulage system design.

---

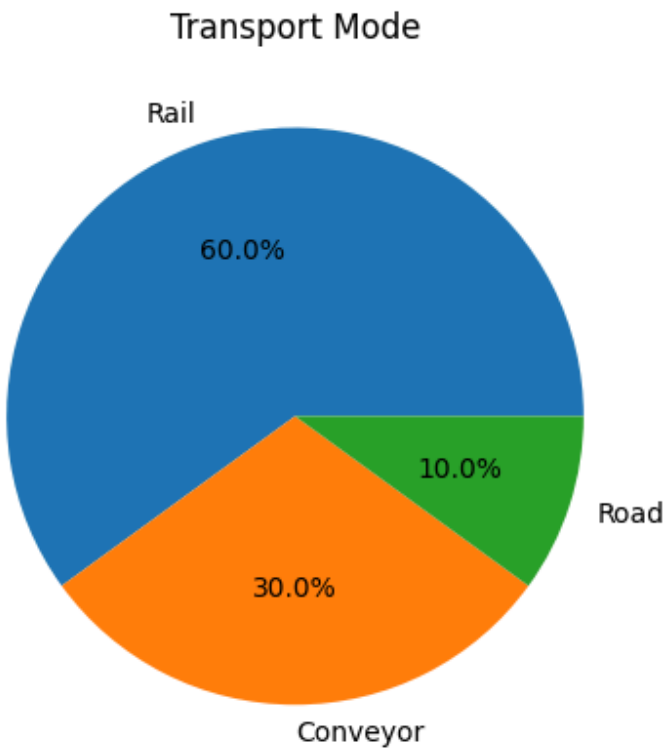
## 📊 5. Mining Method Distribution




Shows dominance of **opencast shovel–dumper method**. (environmentclearance.nic.in)
Modern mechanization ensures high productivity.
**Significance:** Important for understanding mining technology used.

---

## 📊 6. Transportation System




Represents coal transport via **rail + conveyor systems**. (environmentclearance.nic.in)
Bulk transport systems reduce cost and pollution.
**Significance:** Important for logistics efficiency and cost reduction.

---